

AYMAN PATEL

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Professional Summary

Senior Software Engineer with 6+ years of production experience building Java/Spring microservices, event-driven systems, resilient APIs, and cloud-native delivery pipelines. Strong fit for Amadeus' Business Rule Engine: Kafka and message-driven architecture knowledge, Camunda BPMN/DMN rule workflow experience, production troubleshooting with Splunk/Grafana/Dynatrace, and hands-on resilience work with Redis, Resilience4j, circuit breakers, retries, observability, and performance testing.

Core Competencies

Infrastructure: Java 8+, Spring, distributed microservices, event-driven architecture, Kafka, RabbitMQ, REST APIs

Reliability: fault tolerance, circuit breakers, retries, production incident diagnosis, observability, tracing, performance testing

Rule Automation: Camunda, BPMN, DMN, CMMN, workflow automation, event sourcing, configurable business logic

Education

Queen Mary University of London

MSc Advanced Computer Science

London, UK

Sep. 2025 – Sep. 2026

Sardar Vallabhbhai Institute of Technology

Bachelor of Computer Science, GPA: 7.9/10

Gujarat, India

Jul. 2015 – May 2019

Work Experience

Mastercard

Senior Software Engineer

Vadodara, India

Mar. 2024 – Aug. 2025

- Developed backend APIs with Spring, Spring Data JPA, and Hibernate across distributed microservices, applying API Gateway and Event Sourcing patterns for reliable service integration.
- Owned implementation of Resilience4j Circuit Breaker and Retry patterns across 5+ microservices, reducing cascading failures and improving system availability by 15%.
- Owned production incident diagnosis using Splunk, Grafana, and Dynatrace, improving troubleshooting of mission-critical customer-facing services and reducing mean time to resolution.
- Owned frontend observability using Grafana and OpenTelemetry to surface client-side performance and traceability insights for production workflows.
- Owned load and performance testing with k6 and InfluxDB to track throughput trends, regressions, and reliability under traffic.
- Introduced Redis as a cache layer for Mastercard APIs to optimize response times for frequently accessed data.
- Leveraged PostgreSQL JSONB to build flexible, schema-free dynamic form handling for configurable product workflows.
- Translated product manager requirements into technical specifications, user stories, and acceptance criteria for cross-functional engineering teams.

Mastercard

Software Engineer II

Vadodara, India

Mar. 2022 – Feb. 2024

- Built backend APIs for Legal solutions using Spring Boot, Oracle and Hibernate as ORM
- Owned building greenfield frontend applications in React, Redux, JavaScript, and Spring; served as a core maintainer for performance, code quality, and accessibility standards.
- Built Gradle pipelines to streamline dependency management, artifact generation, and release readiness.
- Wrote unit tests with Jest and React Testing Library plus end-to-end tests with Playwright and Cypress, reaching 95% test coverage and reducing CI e2e runtime from 8 hours to 3 hours.
- Improved UI load performance by 30% by migrating delivery to an Akamai CDN.

Mastercard

Software Engineer I

Vadodara, India

Jul. 2019 – Feb. 2022

- Designed and implemented RESTful APIs to automate BPM workflows using Camunda in enterprise payment systems.
- Modelled complex business logic using BPMN, DMN, and CMMN standards, directly relevant to configurable rule evaluation and automated operational decisions.
- Designed CI/CD pipelines to deploy Angular.js applications using Grunt.js and Jenkins.
- Developed and maintained Hardware Security Module integration for mobile encryption using MDES tokenisation, securing card data transmission across payment flows.

Nebius Academy

AI Engineering Fellow

London, UK

Mar. 2025 – May 2025

- Deployed production-ready LLM inference with vLLM, Docker, and Kubernetes on Nvidia H100 infrastructure using Nebius cloud.
- Built AI agents using LLM APIs and MCPs with reasoning pipelines, RAG for factual retrieval, and production-oriented evaluation loops.
- Studied transformer architectures, attention mechanisms, and inference optimization including KV-cache, MoE, RoPE, and LoRA fine-tuning.

Personal Projects

Nebius Project – Repository Summarizer

- Built a FastAPI and LLM service that generates human-readable GitHub repository summaries, extracts code skeletons via AST and Tree-sitter, prioritizes source files, and assembles context within a 7000-token budget.

Java CLI using GraalVM

- Built a Java CLI tool with Picocli to query the Stack Overflow API, then compiled it to a native binary with GraalVM for zero-JVM-startup execution.

Go Microservices

- Built an end-to-end cloud-native microservice in Go with service discovery, distributed tracing, and containerized deployment.

Hackathons

Encode Hackathon

London, UK – 3rd place

- Built Phantom Fraud, an AI-powered financial risk engine for Solana using Solana CLI, OpenRouter, and Helius APIs to detect wash-trading patterns.

AI Hackathon

Vadodara, India – Mar. 2025 – 3rd place

- Built a code-review assistant powered by fine-tuned Gemini Pro Vision with a PostgreSQL-backed developer UI.

AI Hackathon

Vadodara, India – Sep. 2024 – 2nd place

- Developed RAG-based internal documentation search using Langchain and Chroma vector database.

Technology Hackathon

Vadodara, India – 2nd place

- Migrated a React application to Next.js, improving page performance.

Writing

Personal Blog

40+ articles

- Published 40+ technical articles on system design and architecture, with work featured in the eBPF newsletter.
- Wrote infrastructure and backend pieces on eBPF, microfrontends, change data capture, Grafana, Redis, infrastructure-as-code, Localstack, RAG, and LLM observability.

Engineering Whitepapers

Mastercard

- Produced company-wide whitepapers on Accessibility and Microfrontend architecture; earned CTO recognition, adoption as internal reference material, and circulation to 200 engineers.

Talks / Seminars

Kafka Ecosystem Seminar

Oct. 2018

- Delivered a technical seminar covering Kafka Architecture, Kafka Streams, Kafka Connect, and AMQP messaging patterns.

Git and Version Control Systems Seminar

- Presented an introductory talk on Git workflows and version control best practices for undergraduate students.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, Go

Backend: Spring, Spring Data JPA, Hibernate, REST APIs, Microservices, Event Sourcing, CQRS, FastAPI, Django

Messaging and Data: Kafka, RabbitMQ, PostgreSQL, Oracle, MongoDB, Redis, Debezium, Change Data Capture

Cloud and DevOps: Docker, Kubernetes, AWS, Terraform, Localstack, Ansible, PCF, Jenkins, Gradle

Reliability and Observability: Grafana, Prometheus, Splunk, Dynatrace, OpenTelemetry, k6, InfluxDB, Resilience4j

Workflow Automation: Camunda, BPMN, DMN, CMMN, rule-driven automation, business process modelling

Testing: JUnit, Mockito, Jest, React Testing Library, Cypress, Playwright, Testcontainers